NERVOUS MACHINE

Agents That Learn.

A learning protocol for AI agents and physical systems.

Agents build certainty about the real world, get curious about what they don't know, and share discoveries across trusted networks.

Heidi Bennett, Founder

heidi@nervousmachine.com • nervousmachine.com

The Gap That Won't Close

~20%

Persistent sim-to-real gap
that hasn't closed across releases

- AI adopted at unprecedented scale with low trust and reliability
- Foundation models compress individual variance; the signal needed for real-world deployment
- Agents run siloed workflows where learning doesn't backpropagate
- Satellites lose 30-40% operational capacity to safety margins
- Autonomous systems execute on priors that don’t fit reality

Everything lives at the edge of some distribution. The mean is a myth.

Not Memory. Learning.

Agent Memory

- Stores facts and retrieves them
- All information treated equally
- Static until overwritten
- Isolated per agent
- More data = more noise

Agent Learning

- Builds certainty through evidence
- Knows what it knows and what it doesn't
- Gets curious about gaps in understanding
- Agents learn from each other
- More evidence = sharper world model

The Learning Loop

Same protocol for people, agents, and physical systems.

Observe

Agent encounters
the real world

→

Learn

Certainty rises
or falls with evidence

→

Get Curious

Surfaces what it
doesn't know

→

Share

Other agents
build on it

$$\eta(Z) = \eta_{max} \times (1 - \sigma(k \times (Z - 0.5)))$$

Low certainty = big updates. High certainty = stability. Internal priors evolve into learned certainty from external signals.

What Agents Learn

The same protocol works everywhere. The agent decides what matters.

Research Agents

Run experiments, encode findings, build shared certainty. Independent replication raises confidence. Contradictions lower it.

Learns: Experiment results, hardware-specific outcomes, what to try next

Physical Systems

Satellites, robots, and sensors learn the physics of their environment. Error signals from reality close the sim-to-real gap at the edge.

Learns: Atmospheric density, thermal dynamics, causal corrections (1KB vectors)

Personal Agents

Build evolving certainty about preferences and expertise. Corrections refine the model, not replace it.

Learns: Communication style, domain expertise, decision patterns

THE TRIANGLE

One protocol. Three node types. Infinite connections.

PERSONAL PODS

Your pod is yours. No PII leaves. Your context travels across every machine you touch.

Connect to a Waymo, trust handshake, ride personalized without signup.

DEVICE PODS

Every satellite, robot, or sensor learns and carries context shareable to trusted networks. Raw telemetry never leaves.

"Density at this regime is 23% higher than models predict, certainty 0.74"

AGENT PODS

Structured context for the agent-to-agent economy. Identity and trust that other agents can plug into.

AI-powered VC agents tracking certainty across investment diligence.

Privacy by Design

Privacy isn't a feature. It's what makes collective learning possible.

STAYS IN YOUR POD

- Raw data & telemetry
- Personal identifiers (PII)
- Hardware specs & IP
- Proprietary maneuvers

PROPAGATES TO NETWORK

- Causal relationships
- Uncertainty bounds
- Anonymized patterns
- Validated corrections

"People with expertise X tend to experience Y" propagates. Your identity doesn't.

Two Live Prototypes

Same architecture. Opposite ends of the spectrum.

Agent Learning Beta

AI learns about people, research, and decisions

- MCP server for model-agnostic integration
- 39 tools for certainty tracking & learning
- Adaptive learning rate η(Z)
- Curiosity triggers for unknowns
- 20 beta users including AI-powered VC
- Karpathy autoresearch integration (23K stars)

Space Atlas

AI learns about physics

- Live dashboard: space-atlas.nervousmachine.com
- 150 voxels mapped in LEO
- 680 causal edges
- 75 voxels at certainty > 0.70
- 6,000+ debris objects as sensors
- GRACE-FO corridor: >.91 certainty all voxels

Personal context and physical context — unified by causal vectors.

Why Foundation Models Need This

But won't build it themselves.

Wrong Paradigm

They think in epochs and loss functions, not real-time certainty updates from error signals.

No Data Moat

They don't have distributed sensors generating validation data from satellites, factories, and humans.

Neutrality Problem

If Google owns the protocol, no one uses it. If Anthropic owns it, same. Must be neutral.

Resource Pressure

Energy backlash, scaling ceilings, safety scrutiny. Edge learning at 2-5W beats retraining at megawatts.

The interoperability layer must be a protocol, not a product owned by a model company.

Business Model

Personal Pods	**Free / Low-cost**	User owns MongoDB instance. We never see data. Drives adoption.
Enterprise Pods	**Subscription**	Cross-team learning. The pod stays when employees leave.
Device Pods	**Licensing**	Robotics, satellites, manufacturing. Self-improving physical systems.
FM Calibration	**Partnership**	Certainty-quantified priors + discovery of unknowns. 100x more efficient than retraining.

Network effects compound: more pods = better collective learning for everyone.

Traction & Validation

2	20	4
Live Prototypes	Beta Users	Provisional Patents Filed

Organic Engagement (no paid marketing, no intros):

- NASA, Space Force Researchers
- Dan Goldin (former NASA Administrator)
- Andrew Trask (DeepMind) — signal-boosted RL/error-signal research
- Chief Engineers at Boeing, Lockheed Martin, SpaceX, Blue Origin, Nvidia
- Tribute Labs (AI-powered VC) — signed up for beta, exploring integration + investment
- Karpathy autoresearch (23K stars, 2.9K forks) — PR submitted for distributed validation

The world is full of edge cases.

We're building intelligence that learns from them.

Two prototypes live. The protocol is real. Every node gets curious.

context.nervousmachine.com

heidi@nervousmachine.com